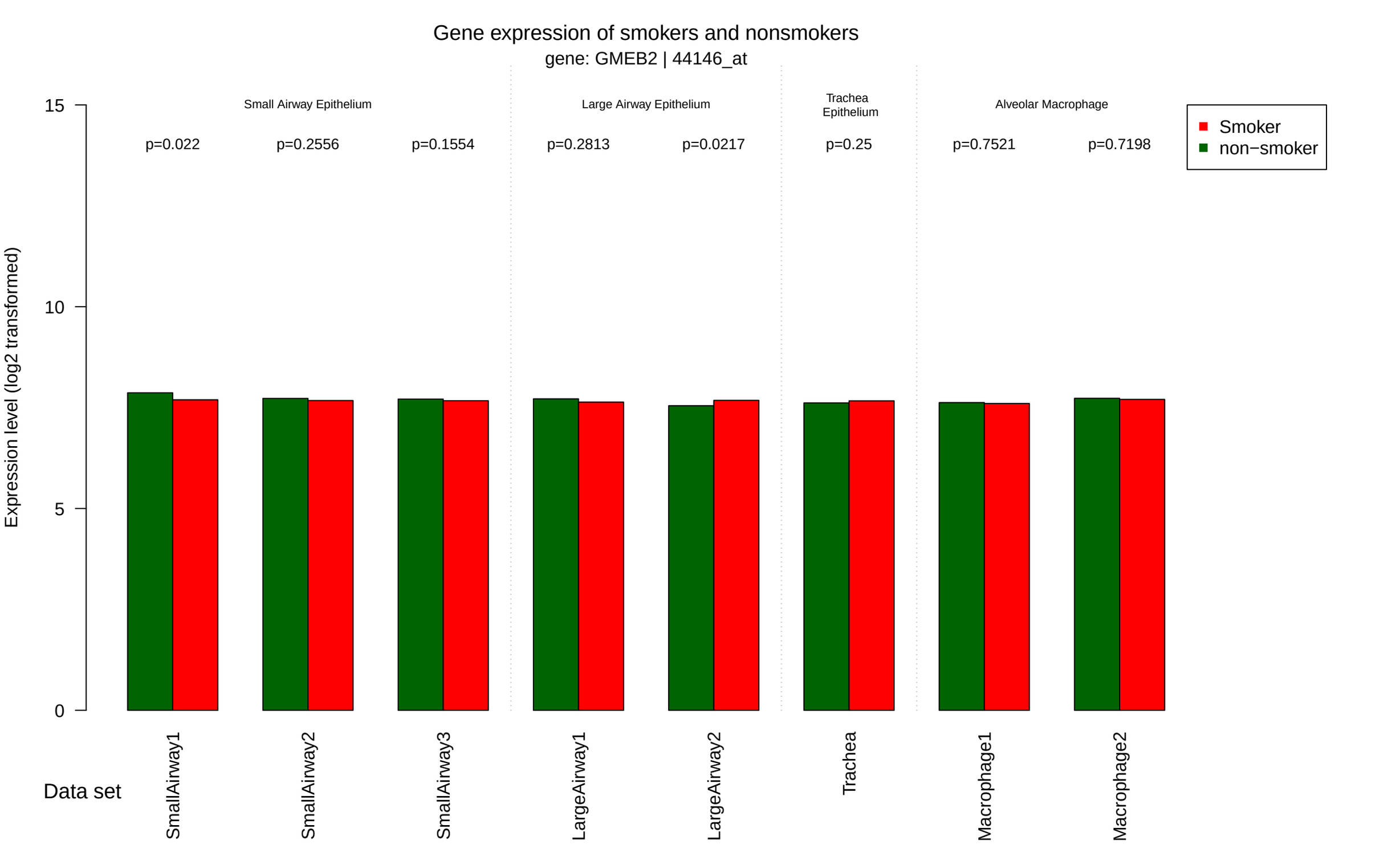
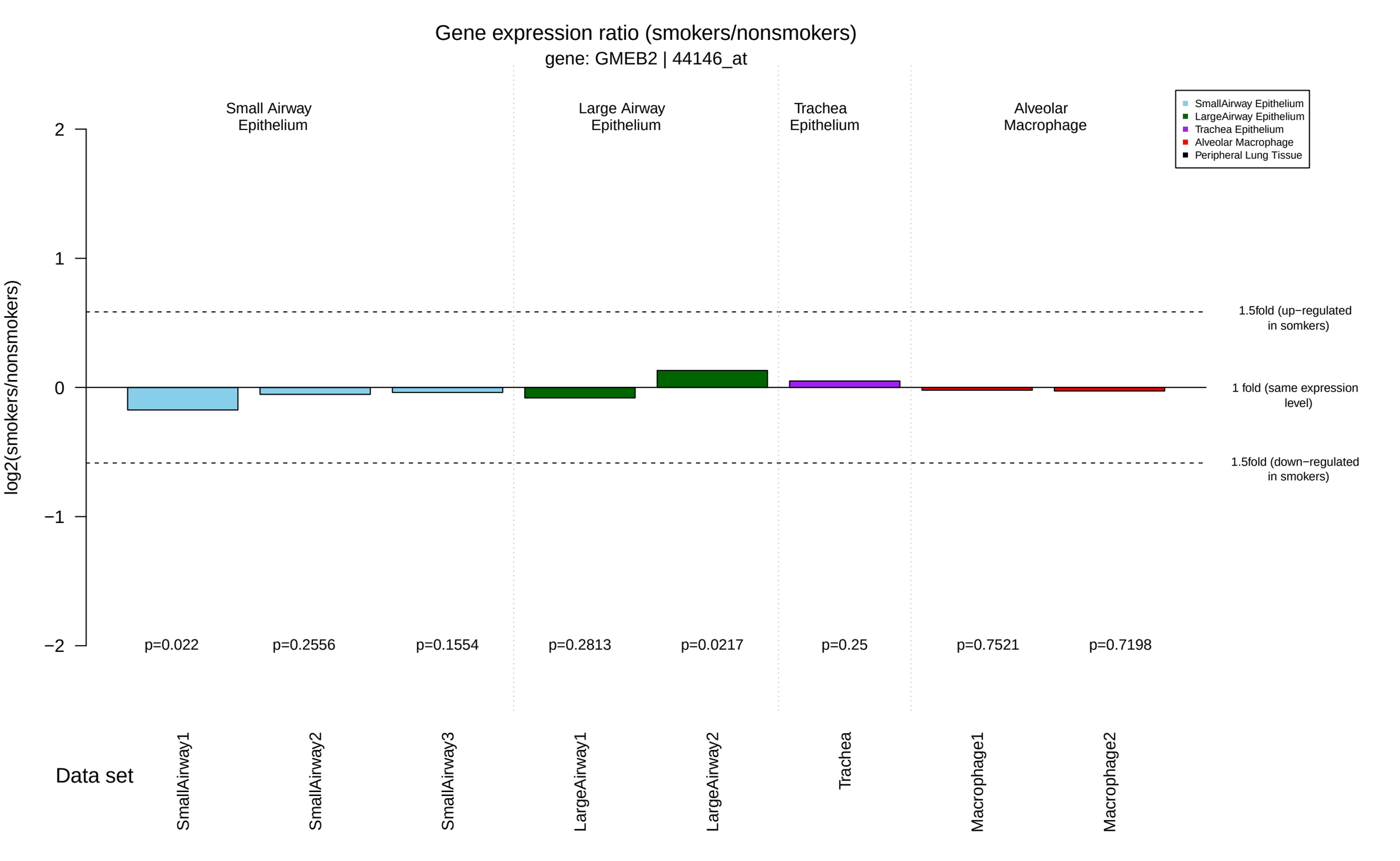
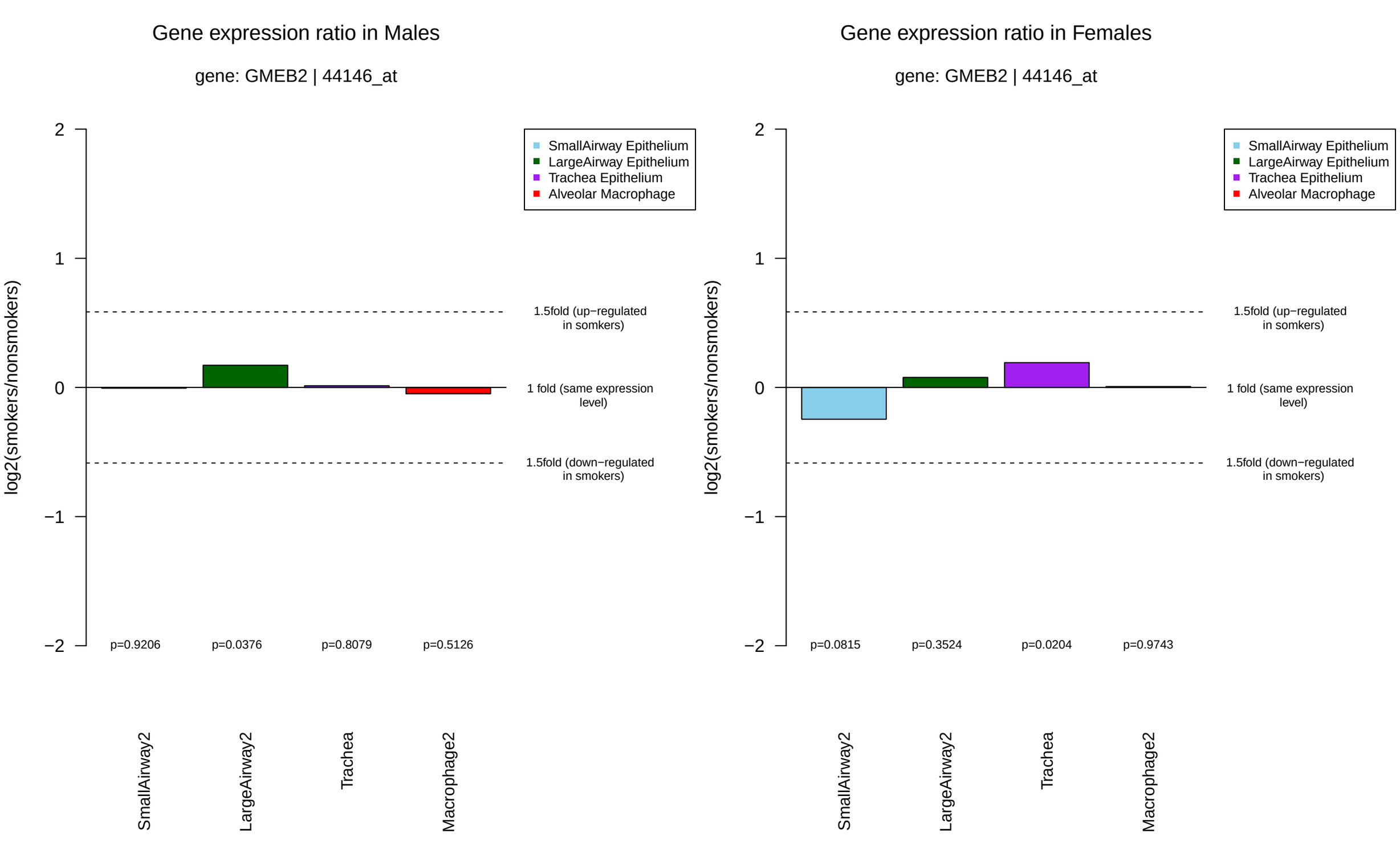
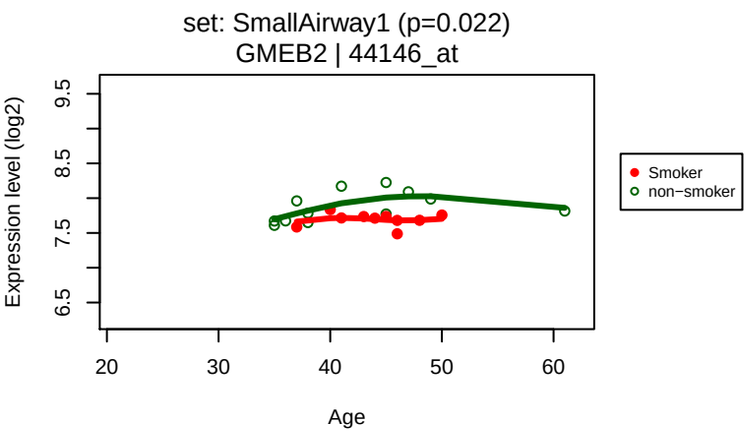
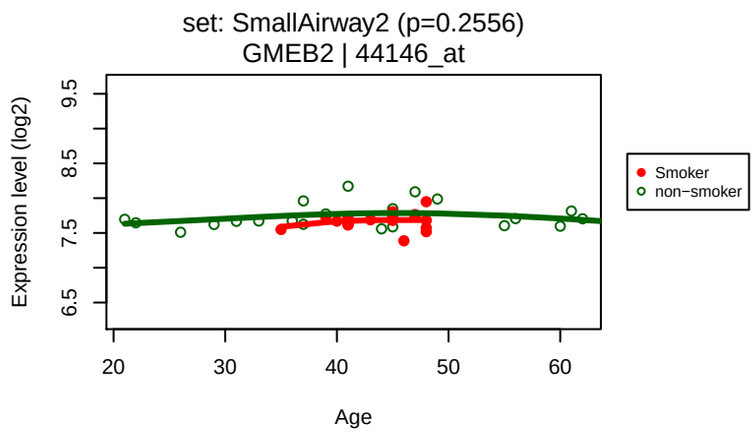
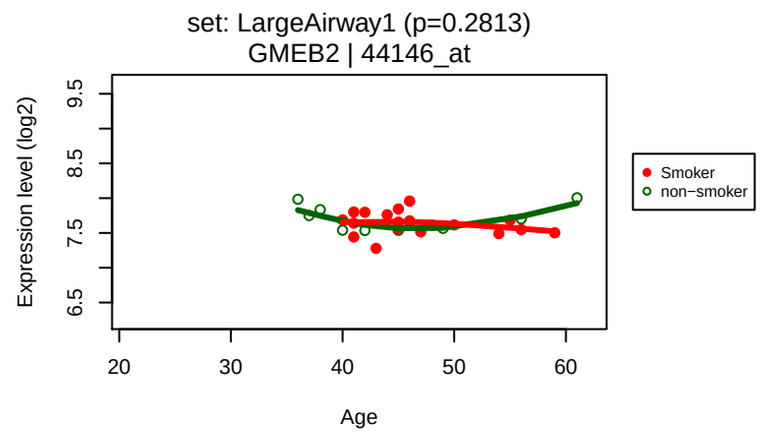

Gene expression of smokers and nonsmokers
gene: GMEB2 | 44146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.022
p=0.2556
p=0.1554
p=0.2813
p=0.0217
p=0.25
p=0.7521
p=0.7198
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

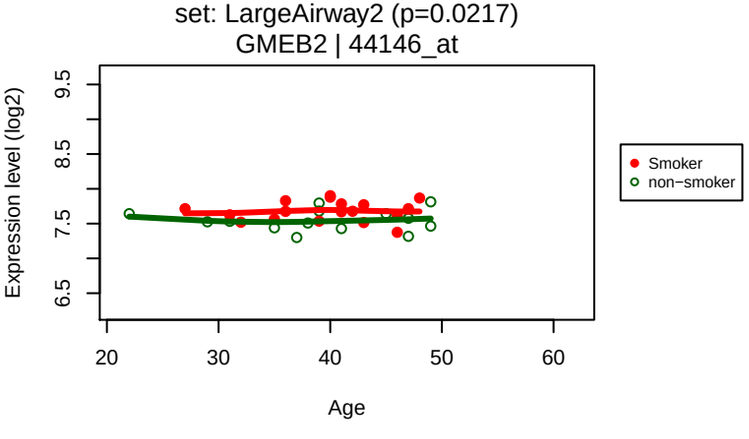

Gene expression ratio (smokers/nonsmokers)
gene: GMEB2 | 44146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.022
p=0.2556
p=0.1554
p=0.2813
p=0.0217
p=0.25
p=0.7521
p=0.7198
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

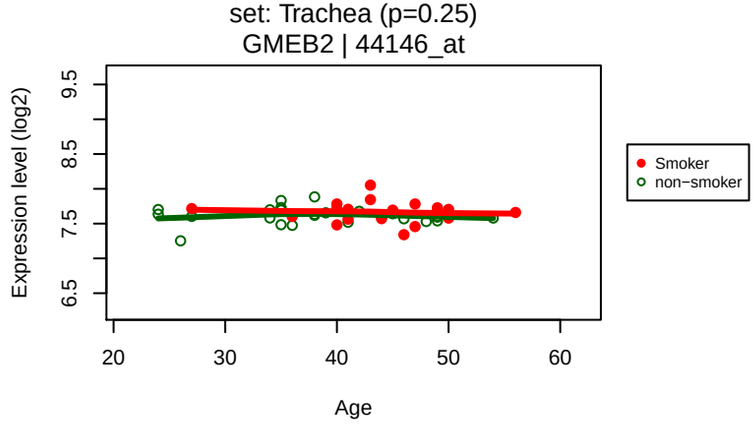

Gene expression ratio in Males
gene: GMEB2 | 44146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9206
p=0.0376
p=0.8079
p=0.5126
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GMEB2 | 44146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0815
p=0.3524
p=0.0204
p=0.9743
SmallAirway2
LargeAirway2
Trachea
Macrophage2

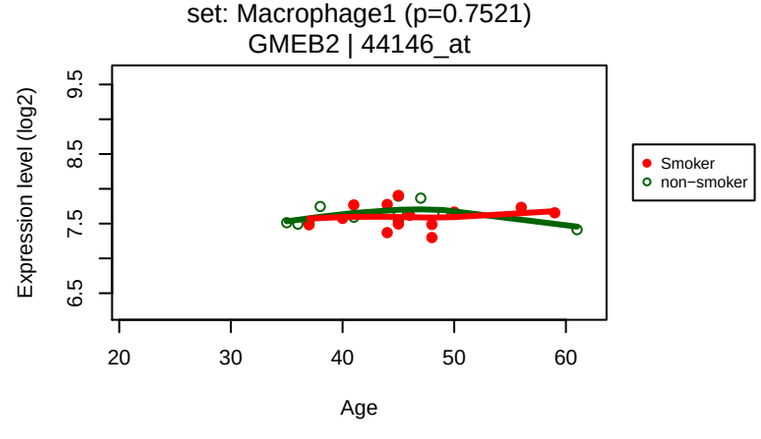

set: SmallAirway1 (p=0.022)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

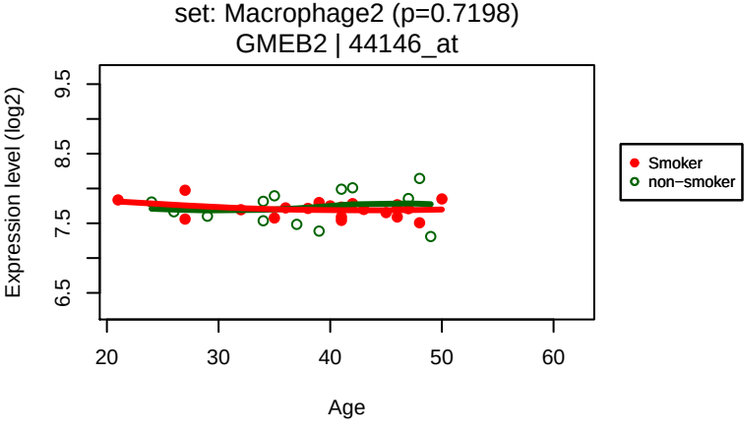

set: SmallAirway2 (p=0.2556)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

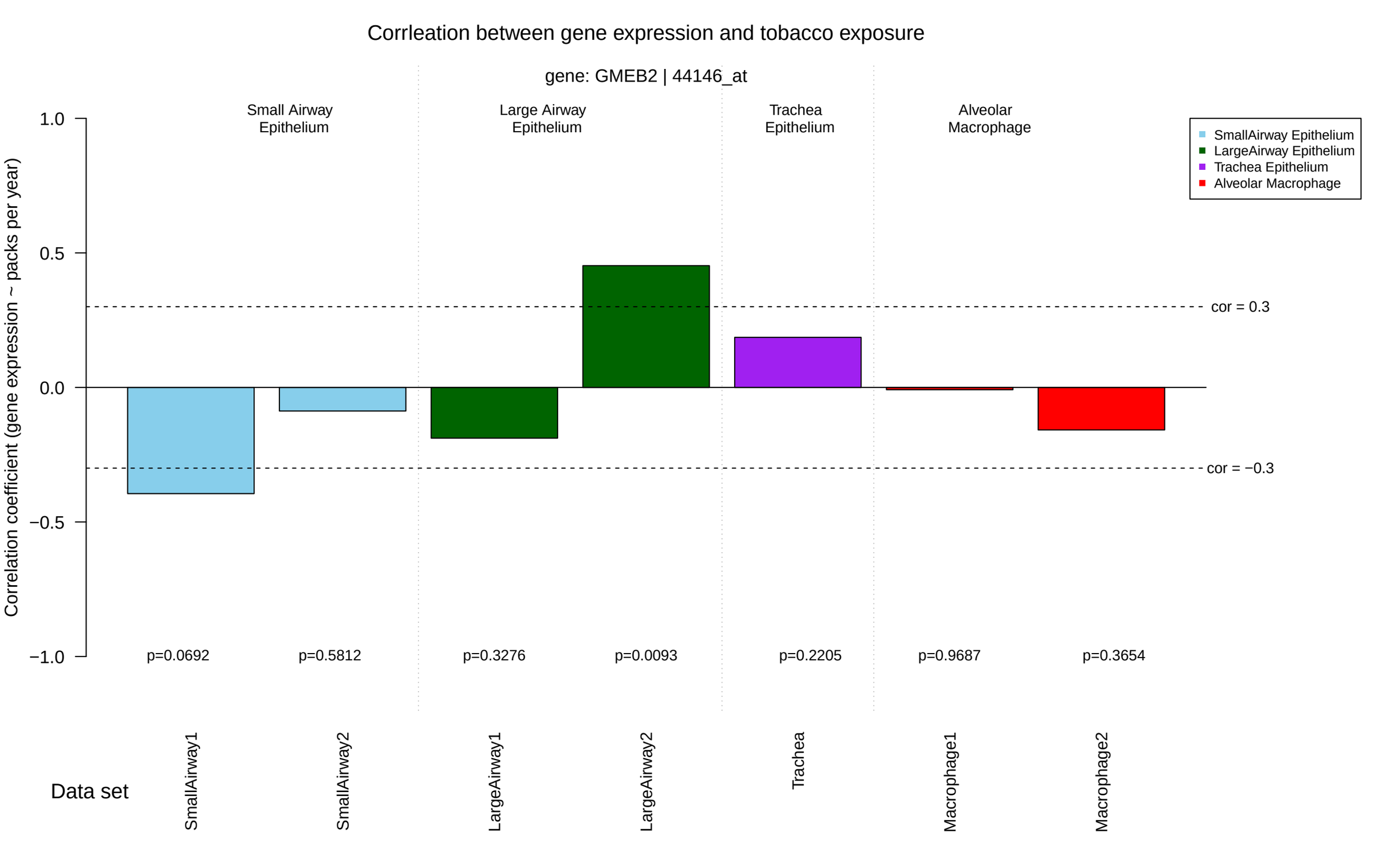

set: LargeAirway1 (p=0.2813)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0217)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.25)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7521)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7198)
GMEB2 | 44146_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GMEB2 | 44146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0692
p=0.5812
p=0.3276
p=0.0093
p=0.2205
p=0.9687
p=0.3654
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2